Middle East Technical University
Department of Electrical and Electronics Engineering
EE301: Signals and Systems I

Solutions for Homework 3
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1. The Fourier series representation of $x_p(t)$ is

$$x_p(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\frac{2\pi}{T_0}t} = a_0 + \sum_{k=1}^{\infty}\left(a_k\, e^{jk\frac{2\pi}{T_0}t} + a_{-k}\, e^{-jk\frac{2\pi}{T_0}t}\right)$$

Owing to be a real-valued signal, we know that its Fourier series coefficients have a form such that $a_k^* = a_{-k}$. Thus, we can write

$$x_p(t) = a_0 + \sum_{k=1}^{\infty}\left(a_k\, e^{jk\frac{2\pi}{T_0}t} + a_k^*\, e^{-jk\frac{2\pi}{T_0}t}\right) = a_0 + \sum_{k=1}^{\infty} 2\mathbf{Re}\left\{a_k\, e^{jk\frac{2\pi}{T_0}t}\right\}$$

Let $a_k = \rho_k e^{j\theta_k}$, where $\rho_k = |a_k|$ and $\theta_k = \angle a_k$, then we get

$$x_p(t) = a_0 + \sum_{k=1}^{\infty} 2\mathbf{Re}\left\{\rho_k e^{j\theta_k}\, e^{jk\frac{2\pi}{T_0}t}\right\} = a_0 + \sum_{k=1}^{\infty} 2\rho_k \cos\left(k\frac{2\pi}{T_0}t + \theta_k\right)$$

If we express $x_p(t)$ in the form:

$$x_p(t) = \sum_{k=0}^{\infty} A_k \cos\left(\omega_k t + \phi_k\right)$$

we find the corresponding parameters as follows:

$$\omega_k = k\tfrac{2\pi}{T_0}, \quad k = 0, 1, 2, \ldots \qquad A_k = \begin{cases} \rho_k, & k = 0 \\ 2\rho_k, & k = 1, 2, \ldots \end{cases} \qquad \phi_k = \theta_k, \quad k = 0, 1, 2, \ldots$$

For $T_0 = 7$ and the given non-zero coefficients, $a_0 = 1$, $a_1 = a_{-1}^* = 1 + j = \sqrt{2}\, e^{j\frac{\pi}{4}}$ and $a_3 = a_{-3}^* = -2 = 2\, e^{j\pi}$, the parameters are

$$\begin{array}{lll} \omega_0 = 0 & A_0 = 1 & \phi_0 = 0 \\ \omega_1 = \dfrac{2\pi}{7} & A_1 = 2\sqrt{2} & \phi_1 = \dfrac{\pi}{4} \\ \omega_3 = \dfrac{6\pi}{7} & A_3 = 4 & \phi_3 = \pi \end{array}$$

2. (a) The coefficient a_0 is the average value of $x_p(t)$ over one period.

$$a_0 = \frac{1}{T}\int_{\langle T\rangle} x_p(t)\,\mathrm{d}t = \frac{1}{4}\left[\int_0^1 \mathrm{d}t + \int_1^2 (2-t)\,\mathrm{d}t\right] = \frac{1}{4}\left[t\Big|_0^1 + \left(2t - \frac{t^2}{2}\right)\Big|_1^2\right] = \frac{1}{4}\left(1 + 2 - \frac{3}{2}\right) = \frac{3}{8}$$

(b) The signal $\frac{d}{dt}x_p(t)$ is a periodic extension of $\frac{d}{dt}x(t)$ such that

$$\frac{d}{dt}x_p(t) = \frac{d}{dt}\left(\sum_{k=-\infty}^{\infty} x(t-kT)\right) = \sum_{k=-\infty}^{\infty} \frac{d}{dt}x(t-kT)$$

for

$$\frac{d}{dt}x(t) = \begin{cases} \delta(t), & 0^- < t < 0^+ \\ -1, & 1 \leqslant t \leqslant 2 \\ 0, & \text{elsewhere} \end{cases}$$

and $T = 4$. Then, its Fourier series coefficients for $k \neq 0$ are

$$\begin{aligned}
b_k &= \frac{1}{T}\int_{\langle T\rangle}\left(\frac{d}{dt}x_p(t)\right) e^{-jk\frac{2\pi}{T}t}\,\mathrm{d}t \\
&= \frac{1}{4}\left[\int_{0^-}^{0^+}\delta(t)\,e^{-jk\frac{\pi}{2}t}\,\mathrm{d}t + \int_{1}^{2}(-1)\,e^{-jk\frac{\pi}{2}t}\,\mathrm{d}t\right] \\
&= \frac{1}{4}\left[1 - \frac{1}{jk\frac{\pi}{2}}\left(e^{-jk\frac{\pi}{2}} - e^{-jk\pi}\right)\right] \\
&= \frac{1}{4}\left[1 - \frac{e^{-jk\frac{3\pi}{4}}}{jk\frac{\pi}{2}}\left(e^{jk\frac{\pi}{4}} - e^{-jk\frac{\pi}{4}}\right)\right] \\
&= \frac{1}{4}\left[1 - \frac{2}{k\frac{\pi}{2}}\sin\left(k\frac{\pi}{4}\right)e^{-jk\frac{3\pi}{4}}\right] \\
&= \frac{1}{4} - \frac{1}{k\pi}\sin\left(k\frac{\pi}{4}\right)e^{-jk\frac{3\pi}{4}}
\end{aligned}$$

and it is obvious that $b_0 = 0$ since any constant, namely DC term, will disappear after taking derivative. Let's check it.

$$b_0 = \frac{1}{T}\int_{\langle T\rangle}\left(\frac{d}{dt}x_p(t)\right)\mathrm{d}t = \frac{1}{4}\left[\int_{0^-}^{0^+}\delta(t)\,\mathrm{d}t + \int_{1}^{2}(-1)\,\mathrm{d}t\right] = \frac{1}{4}\left[1 + (-t)\Big|_{1}^{2}\right] = 0$$

Thus, the Fourier series representation of $\frac{d}{dt}x_p(t)$ is

$$\frac{d}{dt}x_p(t) = \sum_{k=-\infty}^{\infty} b_k\,e^{jk\frac{2\pi}{T}t} = \sum_{\substack{k=-\infty \\ k\neq 0}}^{\infty}\left[\frac{1}{4} - \frac{1}{k\pi}\sin\left(k\frac{\pi}{4}\right)e^{-jk\frac{3\pi}{4}}\right]e^{jk\frac{\pi}{2}t}$$

(c) By the differentiation property of the Fourier series, we know that

$$b_k = jk\frac{2\pi}{T}a_k$$

where a_k and b_k are the Fourier series coefficients of $x_p(t)$ and $\frac{d}{dt}x_p(t)$, respectively. To find a_k, we can use the following expression

$$a_k = \frac{T}{jk2\pi}b_k, \quad k \neq 0$$

For $k = 0$, a_0 is found in part (a). Thus, the Fourier series coefficients of $x_p(t)$ are

$$a_k = \begin{cases} \dfrac{3}{8}, & k = 0 \\ \dfrac{1}{jk2\pi} - \dfrac{2}{j(k\pi)^2} \sin\left(k\dfrac{\pi}{4}\right) e^{-jk\frac{3\pi}{4}}, & \text{otherwise} \end{cases}$$

3. (a) The periodic signal $x_p(t)$ has the Fourier series coefficients

$$a_k = \begin{cases} 4, & k = 0 \\ \left[3^k - 3^{-k}\right], & \text{otherwise} \end{cases}$$

- If $a_k = a_{-k}^*$ for all k, then $x_p(t)$ is real.

Let's check it.

$$a_{-k}^* = \left[3^{-k} - 3^k\right] \neq a_k$$

The given signal $x_p(t)$ is **not real** since $a_k \neq a_{-k}^*$.

(b)
- If $a_k = a_{-k}$ for all k, then $x_p(t)$ is even.
- If $a_k = -a_{-k}$ for all k and so $a_0 = 0$, then $x_p(t)$ is odd.

Let's check it.

$$a_{-k} = \left[3^{-k} - 3^k\right] = -\left[3^k - 3^{-k}\right] = -a_k$$

We observe that $x_p(t)$ is not even since $a_k \neq a_{-k}$ for all k. Moreover, $x_p(t)$ is not odd owing to $a_0 \neq 0$. Thus, the given signal $x_p(t)$ is **neither even nor odd**.

(c) By the differentiation property of the Fourier series, the Fourier series coefficients of $\dfrac{d}{dt}x_p(t)$ are

$$b_k = jk\omega_0 a_k = \begin{cases} 0, & k = 0 \\ jk\omega_0\left[3^k - 3^{-k}\right], & \text{otherwise} \end{cases}$$

- If $b_k = b_{-k}$ for all k, then $\dfrac{d}{dt}x_p(t)$ is even.
- If $b_k = -b_{-k}$ for all k, then $\dfrac{d}{dt}x_p(t)$ is odd.

Let's check it.

$$b_{-k} = -jk\omega_0\left[3^{-k} - 3^k\right] = jk\omega_0\left[3^k - 3^{-k}\right] = b_k$$

The given signal $\dfrac{d}{dt}x_p(t)$ is **even** since $b_k = b_{-k}$.

4. (a) The causal CT and LTI system whose input $x(t)$ and output $y(t)$ are related by the following differential equation:

$$\frac{d}{dt}y(t) + 2y(t) = \frac{d}{dt}x(t)$$

Let the Fourier series representation of input and output signals be

$$x(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t}$$

$$y(t) = \sum_{k=-\infty}^{\infty} b_k\, e^{jk\omega_0 t}$$

respectively.

After substituting these expressions in the given differential equation, we obtain

$$\sum_{k=-\infty}^{\infty} (jk\omega_0 + 2)\, b_k\, e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} jk\omega_0\, a_k\, e^{jk\omega_0 t}$$

and therefore

$$b_k = \frac{jk\omega_0}{2 + jk\omega_0}\, a_k$$

For the given input signal,

$$x(t) = \sin \pi t = \frac{1}{j2} e^{j\pi t} - \frac{1}{j2} e^{-j\pi t}$$

the fundamental frequency is

$$\omega_0 = \pi$$

and the non-zero Fourier series coefficients are

$$a_1 = \frac{1}{j2} \qquad a_{-1} = \frac{-1}{j2}$$

Hence, the non-zero Fourier series coefficients of $y(t)$ are

$$b_1 = \frac{j\pi}{2 + j\pi} \times \frac{1}{j2} = \frac{\pi}{4 + j2\pi} \qquad b_{-1} = \frac{-j\pi}{2 - j\pi} \times \frac{-1}{j2} = \frac{\pi}{4 - j2\pi}$$

(b) For the given input signal,

$$x(t) = \cos(2\pi t) + \sin\left(3\pi t + \frac{\pi}{6}\right) = \frac{1}{2} e^{j2\pi t} + \frac{1}{2} e^{-j2\pi t} + \frac{1}{j2} e^{j\frac{\pi}{6}} e^{j3\pi t} - \frac{1}{j2} e^{-j\frac{\pi}{6}} e^{-j3\pi t}$$

the fundamental frequency is

$$\omega_0 = \pi$$

and the non-zero Fourier series coefficients are

$$a_2 = \frac{1}{2} \qquad a_{-2} = \frac{1}{2}$$

$$a_3 = \frac{1}{j2} e^{j\frac{\pi}{6}} \qquad a_{-3} = \frac{-1}{j2} e^{-j\frac{\pi}{6}}$$

Hence, the non-zero Fourier series coefficients of $y(t)$ are

$$b_2 = \frac{j2\pi}{2 + j2\pi} \times \frac{1}{2} = \frac{j\pi}{2 + j2\pi} \qquad b_{-2} = \frac{-j2\pi}{2 - j2\pi} \times \frac{1}{2} = \frac{-j\pi}{2 - j2\pi}$$

$$b_3 = \frac{j3\pi}{2 + j3\pi} \times \frac{1}{j2} e^{j\frac{\pi}{6}} = \frac{3\pi e^{j\frac{\pi}{6}}}{4 + j6\pi} \qquad b_{-3} = \frac{-j3\pi}{2 - j3\pi} \times \frac{-1}{j2} e^{-j\frac{\pi}{6}} = \frac{3\pi e^{-j\frac{\pi}{6}}}{4 - j6\pi}$$

5. The CT periodic signal $x_p(t)$, the input to an LTI system, is given as

$$x_p(t) = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k \cos\left(k\left(\frac{2\pi}{5}\right)t + \left(\frac{\pi}{6}\right)\right)$$

and $h(t)$, the impulse response of the system, satisfies

$$\int_{-\infty}^{\infty} h(t) e^{-j\omega t}\, \mathrm{d}t = \begin{cases} 1, & |\omega| \leqslant W \\ 0, & |\omega| > W \end{cases}$$

The Fourier series representation of $x_p(t)$ is

$$x_p(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} = a_0 + \sum_{k=1}^{\infty} \left(a_k\, e^{jk\omega_0 t} + a_{-k}\, e^{-jk\omega_0 t} \right)$$

To find a_k, we perform some algebraic manipulations on given expression

$$\begin{aligned}\sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k \cos\left(k\left(\frac{2\pi}{5}\right)t + \left(\frac{\pi}{6}\right)\right) &= \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k \left[\frac{1}{2}\left(e^{j\left(k\frac{2\pi}{5}t+\frac{\pi}{6}\right)} + e^{-j\left(k\frac{2\pi}{5}t+\frac{\pi}{6}\right)}\right)\right] \\ &= \sum_{k=1}^{\infty} \left[\left(\frac{1}{2}\right)^{k+1} e^{j\frac{\pi}{6}} e^{jk\frac{2\pi}{5}t} + \left(\frac{1}{2}\right)^{k+1} e^{-j\frac{\pi}{6}} e^{-jk\frac{2\pi}{5}t}\right]\end{aligned}$$

Then, we replace LHS of expression above with the Fourier series representation of $x_p(t)$

$$a_0 + \sum_{k=1}^{\infty} \left(a_k\, e^{jk\omega_0 t} + a_{-k}\, e^{-jk\omega_0 t} \right) = \sum_{k=1}^{\infty} \left[\left(\frac{1}{2}\right)^{k+1} e^{j\frac{\pi}{6}} e^{jk\frac{2\pi}{5}t} + \left(\frac{1}{2}\right)^{k+1} e^{-j\frac{\pi}{6}} e^{-jk\frac{2\pi}{5}t}\right]$$

and obtain the Fourier series coefficients

$$a_k = \begin{cases} \left(\frac{1}{2}\right)^{-k+1} e^{-j\frac{\pi}{6}}, & k < 0 \\ 0, & k = 0 \\ \left(\frac{1}{2}\right)^{k+1} e^{j\frac{\pi}{6}}, & k > 0 \end{cases}$$

Remember that we can also utilize the results from problem 1 to obtain a_k.

The output signal of the system is

$$\begin{aligned} y(t) &= x_p(t) * h(t) \\ &= \int_{-\infty}^{\infty} x_p(t-\tau) h(\tau)\, \mathrm{d}\tau \\ &= \int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0(t-\tau)} h(\tau)\, \mathrm{d}\tau \\ &= \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} \underbrace{\int_{-\infty}^{\infty} e^{-jk\omega_0 \tau} h(\tau)\, \mathrm{d}\tau}_{=\begin{cases} 1, & |k\omega_0| \leqslant W \\ 0, & |k\omega_0| > W \end{cases}} \end{aligned}$$

Thus, the average power per period of $y(t)$ is smaller than that of $x(t)$ for a finite W.

By using Parseval's relation, the average power per period of $x_p(t)$ is

$$\frac{1}{T_0} \int_{\langle T_0 \rangle} |x_p(t)|^2\, \mathrm{d}t = \sum_{k=-\infty}^{\infty} |a_k|^2$$

To have at least 90% of the average power per period of $x_p(t)$ for the output of system, we should find the minimum value of k_0 such that

$$\frac{\sum_{k=-k_0}^{k_0} |a_k|^2}{\sum_{k=-\infty}^{\infty} |a_k|^2} \geqslant \frac{9}{10}$$

We find the denominator of LHS as below

$$\sum_{k=-\infty}^{\infty} |a_k|^2 = \sum_{k=1}^{\infty} \left(|a_k|^2 + |a_{-k}|^2\right) = \sum_{k=1}^{\infty} \left[\left(\frac{1}{4}\right)^{k+1} + \left(\frac{1}{4}\right)^{k+1}\right] \overset{k=\ell+1}{=} \frac{1}{8}\sum_{\ell=0}^{\infty}\left(\frac{1}{4}\right)^{\ell} = \frac{1}{8}\times\frac{1}{1-\frac{1}{4}} = \frac{1}{6}$$

Similarly, the numerator of LHS can be expressed as

$$\sum_{k=-k_0}^{k_0} |a_k|^2 = \frac{1}{8}\sum_{\ell=0}^{k_0-1}\left(\frac{1}{4}\right)^{\ell} = \frac{1}{8}\times\frac{1-\left(\frac{1}{4}\right)^{k_0}}{1-\frac{1}{4}} = \frac{1}{6}\left[1-\left(\frac{1}{4}\right)^{k_0}\right]$$

Then, we obtain

$$1-\left(\frac{1}{4}\right)^{k_0} \geqslant \frac{9}{10}$$

and the minimum value of k_0 is 2.
Therefore we require

$$W \geqslant k_0\omega_0 = 2\times\frac{2\pi}{5} = 0.8\pi$$

to observe the 1st and 2nd harmonic components of the input signal for the output of system and conserve at least 90% of the average power per period of $x_p(t)$. The minimum value of W satisfying this condition is 0.8π.

6.

- From information 1 and 2, we know that $x(t)$ is a real signal and periodic with $T = 4$. The Fourier series coefficients of $x(t)$ are a_k having a form such that $a_k = a_{-k}^*$.

- From information 3, we can express $x(t)$ as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t} = a_{-2}e^{-j\pi t} + a_{-1}e^{-j\frac{\pi}{2}t} + a_1 e^{j\frac{\pi}{2}t} + a_2 e^{j\pi t} = A_1\cos\left(\frac{\pi}{2}t+\phi_1\right) + A_2\cos\left(\pi t+\phi_2\right)$$

for $a_k = a_{-k}^* = \dfrac{A_k}{2}e^{j\phi_k}$.

- From information 4,

$$-x(t-2) = -A_1\cos\left(\frac{\pi}{2}t-\pi+\phi_1\right) - A_2\cos\left(\pi t-2\pi+\phi_2\right) = A_1\cos\left(\frac{\pi}{2}t+\phi_1\right) - A_2\cos\left(\pi t+\phi_2\right)$$

It is obvious that A_2 should be 0, and therefore $a_2 = a_{-2} = 0$, to satisfy $x(t) = -x(t-2)$. Hence, $x(t)$ should be in the form $A\cos(Bt+C)$ such that

$$x(t) = A_1\cos\left(\frac{\pi}{2}t+\phi_1\right)$$

- From information 5 and Parseval's relation,

$$\underbrace{\frac{1}{4}\int_{-2}^{2}|x(t)|^2\,\mathrm{d}t}_{=2} = \sum_{k=-\infty}^{\infty}|a_k|^2 = 2|a_1|^2 \implies |a_1| = \frac{1}{2}$$

- From information 6, $a_1 = \dfrac{1}{2}$ and

$$x(t) = \cos\left(\frac{\pi}{2}t\right)$$

7. To find the impulse response of the system with the frequency response

$$H(\omega) = \frac{e^{-j6\omega}}{3 + j\frac{\omega-2}{5}}$$

we perform some algebraic manipulations on $H(\omega)$ such that

$$H(\omega) = \frac{e^{-j6\omega}}{3 + j\frac{\omega-2}{5}} = \frac{e^{-j6\omega}}{\frac{1}{5}\left(15 + j\omega - j2\right)} = \frac{5e^{-j6\omega}}{\underbrace{(15 - j2)}_{a} + j\omega} = \frac{5e^{-j6\omega}}{a + j\omega}$$

Since $\mathbf{Re}\{a\} = 15 > 0$, we can use one of the basic Fourier transform pairs

$$e^{-at}u(t) \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad \frac{1}{a + j\omega}$$

By the properties of the Fourier transform, we get

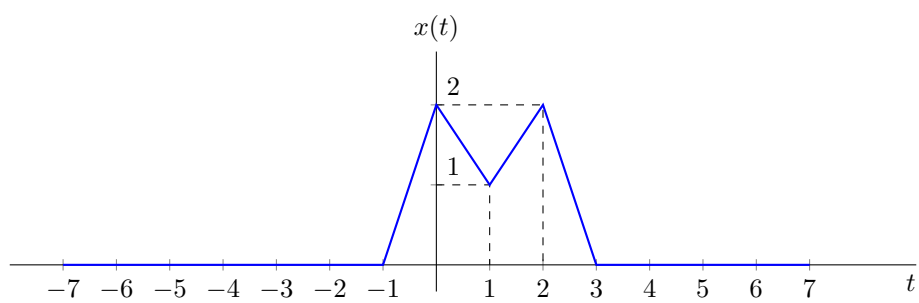
$$5e^{-at}u(t) \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad \frac{5}{a + j\omega} \quad \text{(Linearity)}$$

$$5e^{-a(t-6)}u(t-6) \quad \overset{\mathcal{F}}{\longleftrightarrow} \quad \frac{5e^{-j6\omega}}{a + j\omega} \quad \text{(Time shifting)}$$

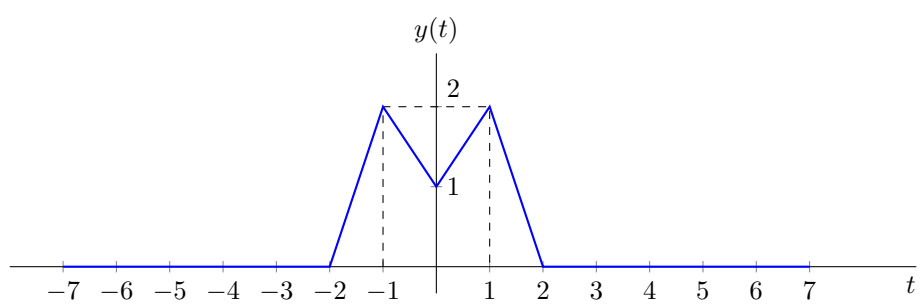
After substituting a, the impulse response of the system is

$$h(t) = 5e^{-(15-j2)(t-6)}u(t-6)$$

8. (a)



Let $y(t) = x(t+1)$, then note that $y(t)$ is real and even as shown below.



By the time shifting property of the Fourier transform

$$y(t) = x(t+1) \overset{\mathcal{F}}{\longleftrightarrow} Y(\omega) = e^{j\omega}X(\omega)$$

Let $X(\omega) = A(\omega)e^{j\Theta(\omega)}$, then

$$\angle Y(\omega) = \omega + \Theta(\omega)$$

We know that $Y(\omega)$ is real and even since $y(t)$ is real and even. Thus, $\angle Y(\omega) = 0$ and $\Theta(\omega) = -\omega$.

(b) By using the Fourier transform equation,

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,\mathrm{d}t$$

we can find $X(0)$

$$X(0) = \int_{-\infty}^{\infty} x(t)\,\mathrm{d}t = 5$$

(c) By using the inverse Fourier transform equation,

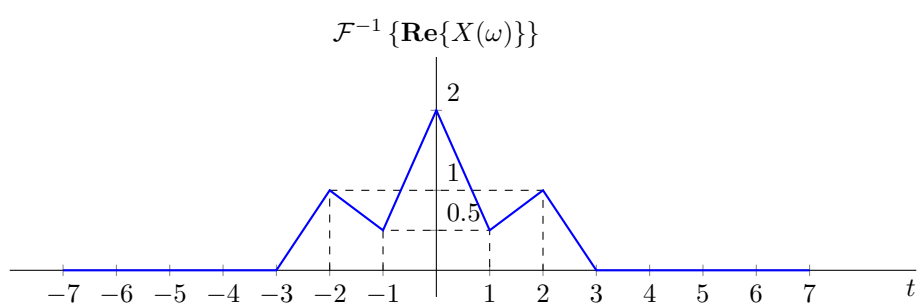
$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,\mathrm{d}\omega$$

we can find the area under $X(\omega)$

$$\int_{-\infty}^{\infty} X(\omega)\,\mathrm{d}\omega = 2\pi x(0) = 4\pi$$

(d) Since $x(t)$ is a real signal, we know that

$$\mathcal{E}\nu(x(t)) \overset{\mathcal{F}}{\longleftrightarrow} \mathbf{Re}\{X(\omega)\}$$

Therefore we get

$$\mathcal{F}^{-1}\left\{\mathbf{Re}\{X(\omega)\}\right\} = \mathcal{E}\nu(x(t)) = \frac{1}{2}\left(x(t) + x(-t)\right)$$